

Wikipedia says...

In the field of computer science, an interface refers to a point of interaction between components, and is applicable at the level of both hardware and software. This allows a component, whether a piece of hardware such as a graphics card or a piece of software such as an Internet browser, to function independently while using interfaces to communicate with other components via an input/output system and an associated protocol.

A software interface may refer to a range of different types of interfaces at different 'levels': an operating system may interface with pieces of hardware, applications or programs running on the operating system may need to interact via streams, and in object-oriented programs, objects within an application may need to interact via methods.

Hardware interfaces exist in computing systems between many of the components such as the various buses, storage devices and other I/O devices. A hardware interface is described by the mechanical, electrical and logical signals at the interface and the protocol for sequencing them (sometimes called signalling). A standard interface, such as SCSI, decouples the design and introduction of computing hardware, such as I/O devices, from the design and introduction of other components of a computing system, thereby allowing users and manufacturers great flexibility in the implementation of computing systems.

differentiator in this faster-than-thought computing era.

Some of today's high-speed interfaces include DDR2 and DDR3 in the memory space; PCIe Gen2, USB2 and USB3 in the input/output space; 802.11n in the wireless communications space; and HDMI in the high-definition multimedia space. Even as these are gaining popularity, other newer technologies are brewing in

research labs of companies like Intel.

Let us take a quick look at these and other recent advancements in high-speed interfaces.

Lightning-fast interfaces for thundering memory speed

Random-access memory (RAM) is a critical element that determines the speed, performance and capabilities of any computing platform. As a result, memory technologies continue to evolve at a rapid rate. However, that is not sufficient—the interfaces of these RAM devices must also evolve to keep pace, so that these memory devices can be utilised easily and completely by various computing systems. And we must say that the interfaces are keeping pace with the memory technologies.

Version 3 of the double-data-rate synchronous dynamic random-access memory (DDR3 SDRAM) interface specification is gaining ground, even as its predecessor DDR2 continues to have a wing of followers.

DDR3 is for high-bandwidth applications. It has the ability to transfer data at twice the rate of DDR2. This enables higher bus rates, higher peak rates and a larger memory module size. The one disadvantage, however, is that DDR3 is not backward-compatible with DDR2.

Ethernet-capable SDRAM data rates have also doubled approximately every three years to 1.8 gigatransfer/second (GT/s). At the same time, the power supply voltage has decreased significantly resulting in lower power consumption. As a result, SDRAM signals are now faster and smaller.

There is a bit of recent activity related to memory interfaces for mobile devices, which now use the mobile DDR interface technology. The new low-power DDR2 (LPDDR2) mobile interface technology is also catching up. But some vendors feel even that is inadequate for the requirements of the next wave of devices which will most likely use three-dimensional integrated circuits based on through-silicon vias or TSVs! Technologies like LPDDR2, despite its 4.3Gbps capability, cannot meet the bandwidth requirements of these devices. Hence players such as the Serial Port Memory Technology Consortium (SPMT), the MIPI Alliance that focuses on standards for mobile interfaces and Rambus' XDR are working towards developing new interfaces that would have two to three times more bandwidth than current technologies.

SPMT, for instance, announced a new specification in June that is based on the new 'SerialSwitch' technology, which reportedly delivers four times the bandwidth, pin for pin, at half the power of existing memory solutions, and offers a low-cost migration from parallel to serial memories. It is backward-compatible with existing LPDDR2, and operates in legacy LPDDR2 mode with up to 1.6 Gbps in parallel mode. It is said to switch into overdrive in serial mode with up to 6.4 Gbps of bandwidth per channel. The technology is optimised for mobile and consumer electronics applications



HDMI cord